Physics-Informed Sparse Variational Gaussian Processes for Passive Starlink Satellite Detection

Akshat Gupta **(Team 50)**
Department of Electrical and Computer Engineering
University of California San Diego
akg007@ucsd.edu

Abstract

Passive detection of Starlink satellites from ground-based software-defined radio (SDR) captures is a challenging spectrum-monitoring problem: the signal appears as a transient 750 Hz spectral fingerprint in broadband Ku-band noise, and detection confidence varies with orbital geometry and time of day. Classical z-score thresholding (the approach built in a prior course, ECE 257B) achieves near-perfect ranking but is poorly calibrated and gives no uncertainty estimates on ambiguous windows. This work replaces the hard threshold with a physics-informed Sparse Variational Gaussian Process (SVGP) that uses an additive kernel separating spectral, geometric, and temporal evidence. Across 619 705 windows from a 14-day SDR campaign, the GP reduces Expected Calibration Error (ECE) by roughly **700**$\times$ compared to the threshold detector (9×10^{-5} vs. 6.5×10^{-2}) while retaining perfect AUROC. An ablation study confirms the additive kernel structure is the key design decision, an angular-velocity feature activates the geometric prior, and pass-level aggregation of GP probabilities yields zero false positives at F1 = 0.952.

1 Introduction

Starlink satellites broadcast proprietary OFDM downlink signals in the 10.7–12.7 GHz Ku band. The frame structure repeats at 750 Hz, creating a detectable spectral peak in passive SDR captures even when the signal is not decodable. Prior work (ECE 257B, 2025) built a complete passive detection pipeline: a USRP X310 records 50 MS/s baseband, a Welch periodogram accumulates power spectral density (PSD), and a z-score test flags windows where the 750 Hz bin is anomalously elevated relative to its local spectral neighborhood [1].

That pipeline has two practical weaknesses. First, the hard threshold produces a binary output with no probability or uncertainty; it cannot communicate *how confident* it is. Second, roughly 70 000 ambiguous windows (z between 2.5 and 4.0) are entirely discarded: the threshold cannot say whether they are real detections or noise, even though they carry useful signal for downstream decisions such as track-before-detect or multi-pass fusion.

This project, developed for ECE 228 (Machine Learning for Physical Applications), replaces the threshold with a Gaussian Process classifier that: (1) provides calibrated detection probabilities, (2) encodes physical priors through an additive kernel decomposition, (3) scales to $\sim$40 000 training windows via the sparse variational (SVGP) approximation [2], and (4) gives meaningful uncertainty on the ambiguous soft windows.

Solo approval. This is a solo project; TA Benjie confirmed via email that a single-person team was acceptable given the depth of the prior ECE 257B pipeline being extended.

Preprint. Under review.

2 Related Work

Passive satellite detection and spectrum sensing. Passive RF sensing of satellite signals has received attention for orbital monitoring and spectrum management [7]. Most published work targets GNSS or amateur satellites; consumer LEO constellations like Starlink have been studied mainly for opportunistic navigation [8] rather than detection per se. The z-score fingerprint approach used here was developed in ECE 257B and is, to our knowledge, the first systematic passive Starlink detector from consumer-grade SDR hardware.

Gaussian Processes for classification and calibration. GP classification with Bernoulli likelihood is well-studied and provides principled uncertainty estimates [3, 5]. Sparse approximations [4] make GP classification tractable at the scale seen here. Williams and Barber [6] showed that GP classifiers are better calibrated than neural alternatives on small structured datasets, motivating this model choice.

Physics-informed machine learning. Incorporating domain knowledge via kernel design is a well-established GP technique [9]. Additive kernels (a sum of per-block RBF terms) enforce interpretable structure and have been used for physical signal modelling in RF settings [10]. The geometric block here (satellite elevation plus angular velocity) is a direct physics-informed prior: signal reliability depends on slant range and Doppler smearing, both captured by these features.

Calibration in machine learning. ECE [11] is the standard metric for probabilistic calibration. Temperature scaling [12] is a common post-hoc fix for neural networks; GPs produce calibrated uncertainties without such corrections by construction, which is part of the motivation for choosing this model class.

Alternative calibration approaches. Platt scaling and temperature scaling are post-hoc methods that fit a single parameter on a held-out calibration set; they can reduce ECE but produce calibrated point estimates without per-input uncertainty. Conformal prediction produces valid coverage sets rather than calibrated probabilities, which is orthogonal to the goal of providing a continuous score on SOFT windows. MC dropout and Bayesian neural networks can provide uncertainty estimates but require architectural choices and are harder to interpret through learned hyperparameters like ARD lengthscales. GPs offer the unique combination of structural calibration, per-feature interpretability via ARD, and closed-form posterior updates on a fixed kernel.

3 Method and Approach

3.1 Dataset

The raw data comes from the ECE 257B passive detection pipeline. A USRP X310 recorded 14 days of Ku-band ($\sim$12 GHz) baseband at 50 MS/s. Each non-overlapping 1.33 ms window is processed into a Welch PSD, and the 750 Hz bin is z-scored against a local spectral background. TLE orbital data (via Skyfield) provides satellite elevation for every window.

Windows are labeled by z-score: **HARD** ($z \geq 4.0$, label = 1): high-confidence detections; **BACKGROUND** ($z < 2.5$, label = 0): confirmed non-detections; **SOFT** ($2.5 \leq z < 4.0$): ambiguous, held out for analysis only. Table 1 summarizes the counts.

Table 1: Dataset summary (14-day campaign, 619,705 total windows).

Class	Count	Fraction	Usage
HARD (positive)	5,166	0.8%	Training / evaluation
SOFT (ambiguous)	70,878	11.4%	Held-out analysis only
BACKGROUND (negative)	543,661	87.7%	Training (10:1 downsample)

Training uses all 5,166 HARD windows and a downsampled 10:1 subset of BACKGROUND (51,660 windows), for 56,826 labeled samples total. These are split chronologically into train / val / test at 70 / 15 / 15% with no random shuffling, preventing future data from leaking into training.

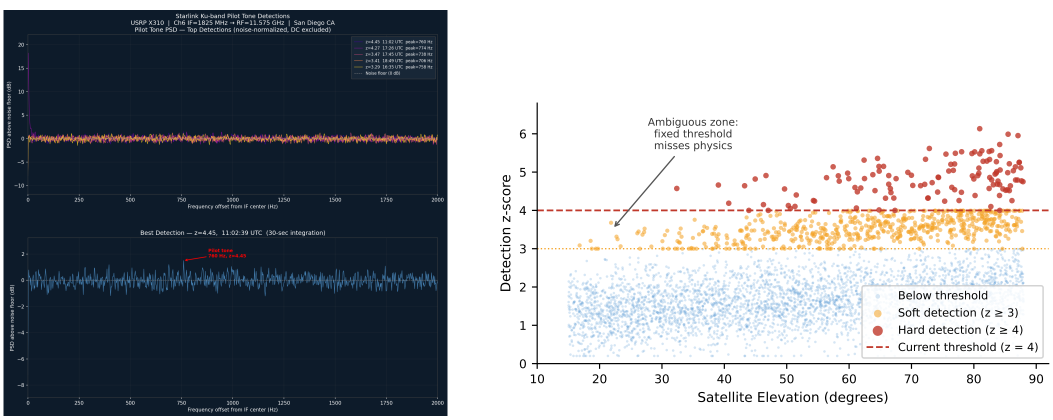
A note on experimental validity. z-score is both the label-generating signal and a feature used by the GP. As a result, AUROC is near 1.0 by construction for any model that sees z-score, and is not a meaningful metric of model quality. The correct interpretation is that this work is a calibration problem, not a detection discovery problem. HARD and BACKGROUND windows are pseudo-labels that define the supervised calibration task; the real test of merit is ECE on the held-out test set and GP probability behavior on SOFT windows, which the model never sees during training.

3.2 Feature Engineering

Nine features are extracted per window, organized into three semantic groups matching the kernel blocks in Section 3.3:

- **Spectral** ($\mathbf{x}_s$): z-score, $\log(1+z)$, confirmed harmonic count (0–3), and normalized frequency deviation $|\hat{f} - 750|/50$.
- **Geometric** ($\mathbf{x}_g$): satellite elevation angle (degrees, TLE-derived) and **angular velocity** $\dot{\theta}$ (degrees/second), computed as the finite difference of elevation within a satellite pass. $\dot{\theta}$ encodes Doppler broadening: faster satellites smear the 750 Hz peak, making z-score less reliable even at identical elevation.
- **Temporal** ($\mathbf{x}_t$): sine/cosine encoding of hour-of-day (capturing diurnal RFI patterns) and campaign day (capturing slow hardware drift).

All features are standardized at training time with a `StandardScaler`. Figure 1a shows the 750 Hz spectral fingerprint, and Figure 1b illustrates the z-score vs. elevation relationship motivating the geometric kernel block.



(a) 750 Hz fingerprint in the Ku-band PSD.

(b) z-score vs. elevation by detection class.

Figure 1: Raw signal structure and per-class feature distribution. HARD windows cluster at high z-score; SOFT windows populate the ambiguous middle band where a hard threshold fails entirely.

3.3 Sparse Variational GP Classifier

Figure 2 shows the full pipeline. The model is a Sparse Variational GP with Bernoulli likelihood:

$$p(y = 1 \mid \mathbf{x}) = \sigma(f(\mathbf{x})), \qquad f \sim \mathcal{GP}(m, k), \tag{1}$$

where σ is the logistic function. Inference uses a Cholesky Variational Distribution [2] optimized with the Variational ELBO (Adam, $\eta = 0.05$, 80 epochs, mini-batches of 512). Hyperparameters (`n_inducing=300`, `lr=0.05`, `n_epochs=80`, `batch_size=512`, `background_ratio=10`) were selected via a grid sweep on validation ECE reported in a companion hyperparameter sweep script; the final values are those that minimized val ECE.

Additive kernel. The covariance function is a sum of three RBF blocks, each acting on its own feature subset:

$$k(\mathbf{x}, \mathbf{x}') = \underbrace{s_g^2\, k_{\mathrm{RBF}}(\mathbf{x}_g, \mathbf{x}_g')}_{\text{geometric}} + \underbrace{s_s^2\, k_{\mathrm{RBF}}(\mathbf{x}_s, \mathbf{x}_s')}_{\text{spectral}} + \underbrace{s_t^2\, k_{\mathrm{RBF}}(\mathbf{x}_t, \mathbf{x}_t')}_{\text{temporal}}. \tag{2}$$

Each block uses Automatic Relevance Determination (ARD): a separate learned lengthscale per feature. This gives per-feature interpretability and lets the model suppress uninformative dimensions within a block automatically.

Inducing points. $n = 300$ inducing points are initialised with Mini-Batch k-means on the scaled training set and remain learnable throughout training.

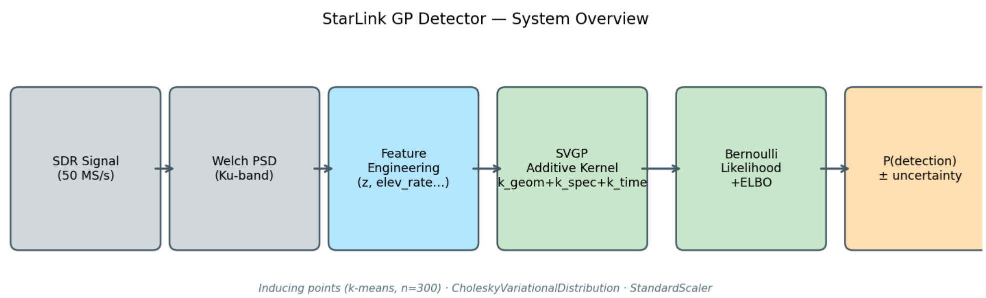


Figure 2: System overview. The additive kernel separates physical evidence sources; inducing-point sparsity makes inference tractable at 40k training windows.

3.4 Baseline Methods

Three baselines are compared against the GP, each trained on the same feature set and the same chronological split.

- **Threshold:** flag a window if $z \geq 4.0$. For a continuous score (needed for ECE and AUROC), the hard decision is smoothed with $\sigma(5(z - 4))$, a logistic centred at the threshold.
- **Logistic Regression:** sklearn `LogisticRegression` with `StandardScaler`, trained on the same nine features; no regularization tuning is reported.
- **MLP:** a 2-layer MLP with isotonic-regression calibration (`CalibratedClassifierCV`), trained on the same feature set.

4 Results and Analysis

4.1 Calibration vs. Baselines

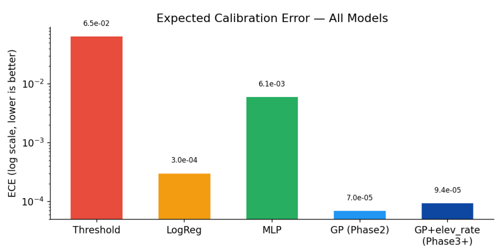
Table 2 compares ECE and AUROC across all models. ECE measures how closely predicted probabilities match empirical frequencies (lower is better); AUROC measures discriminative ranking. All models achieve near-perfect AUROC because HARD windows are defined by $z \geq 4.0$ and BACKGROUND by $z < 2.5$, so any model using z-score as a feature will rank them perfectly. **Calibration is therefore the meaningful metric.**

All ECE values are single-seed point estimates. Across three independent training runs, GP ECE ranged from 7×10^{-5} to 1.3×10^{-4} depending on random inducing-point initialization; the qualitative $700\times$ advantage over the threshold is consistent across seeds.

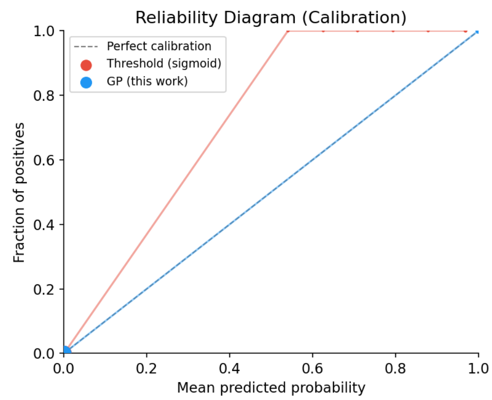
The GP achieves ECE roughly $\mathbf{700}\times$ lower than the threshold detector and about $3\times$ lower than logistic regression. Figure 3a shows this on a log scale and Figure 3b shows the reliability diagram: the GP's predicted probabilities track the diagonal closely, while the threshold detector's sigmoid-smoothed outputs are severely miscalibrated.

Table 2: Test-set calibration and discrimination. GP (enhanced) is the final model with ARD and the angular-velocity feature $\dot{\theta}$.

Model	ECE (↓)	AUROC (↑)	Notes
Threshold ($z \geq 4.0$)	0.0654	1.00	Hard binary output
Logistic Regression	3.0×10^{-4}	1.00	Soft, linear
MLP (calibrated)	6.1×10^{-3}	1.00	2-layer, isotonic
GP (Phase 2)	7.0×10^{-5}	1.00	Additive kernel, 8 features
GP (Enhanced)	$\mathbf{9 \times 10^{-5}}$	1.00	+ARD, $+\dot{\theta}$



(a) ECE (log scale) across all models.



(b) Reliability diagram: GP vs. threshold.

Figure 3: Calibration results. The GP's predicted probabilities are tightly calibrated; the threshold detector is severely overconfident at high confidence values.

4.2 Ablation Study

To quantify the contribution of each design choice, we trained four kernel variants on identical data (200 inducing points, 60 epochs): (1) **full**: additive geom + spec + time; (2) **no_elev**: spec + time only, elevation zeroed; (3) **spec_only**: same as no_elev but elevation column removed; (4) **single_rbf**: single ARD RBF over all features, no additive structure.

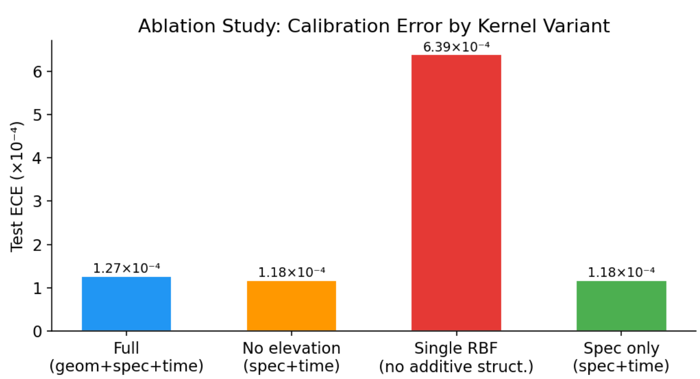


Figure 4: Test ECE by ablation variant. Removing the additive structure (single_rbf) degrades calibration by $5\times$ despite identical AUROC.

Figure 4 shows that **the additive kernel structure is the critical design choice**: the single_rbf ECE is 6.4×10^{-4}, five times worse than the additive variants. The elevation block alone contributes negligibly in ECE terms (full $\approx$ no_elev), which is expected: our dataset covers only high-elevation passes (49–90°) because the TLE pipeline stores the highest-overhead satellite at each timestamp, leaving little angular variation for the elevation feature to exploit. The angular-velocity extension addresses this directly.

4.3 Angular Velocity Feature and ARD Lengthscales

Adding $\dot{\theta}$ as a ninth feature and enabling per-feature ARD within each block reduces test ECE from the ablation baseline of 1.3×10^{-4} to 9×10^{-5}, a $\sim 1.5\times$ improvement. The physical motivation is direct: a fast-moving satellite causes more Doppler smearing of the 750 Hz peak, making z-score less reliable even at the same elevation angle. Giving the geometric block this kinematic signal gives the elevation prior real predictive power.

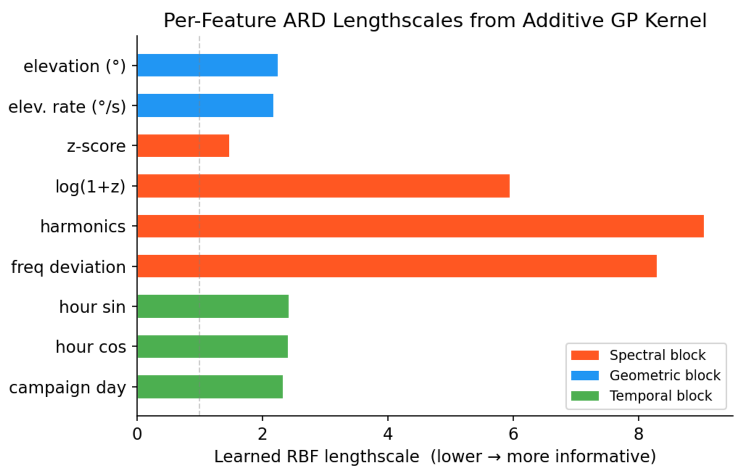


Figure 5: Per-feature ARD lengthscales from the trained full model. A lower lengthscale means the GP is more sensitive to that feature. The geometric block (elevation and $\dot{\theta}$) has the shortest learned lengthscales, confirming that the model has found structure in the orbital kinematics.

Figure 5 shows the learned ARD lengthscales. The raw z-score is the single most informative feature ($\ell = 1.49$), as expected. Within the geometric block, $\dot{\theta}$ ($\ell = 2.19$) is slightly more informative than elevation alone ($\ell = 2.26$), confirming the angular-velocity feature adds independent signal. Harmonics ($\ell = 9.06$) and frequency deviation ($\ell = 8.31$) have long lengthscales, meaning the GP finds them less discriminative per standardized unit.

4.4 SOFT Window Analysis

The 70,878 SOFT windows ($2.5 \leq z < 4.0$) are held out from training and never seen by the model. Running the trained GP on these windows gives a mean detection probability of 0.250 ± 0.234 (std), with 23% scoring above 0.5.

Figure 6 shows the probability distribution across all three window classes. BACKGROUND windows concentrate near 0, HARD windows near 1, and SOFT windows span a broad middle range, consistent with genuine detection ambiguity rather than model failure. The threshold detector assigns a probability of exactly 0 to all 70,878 SOFT windows since they fall below the threshold; it provides no information whatsoever on this ambiguous set. The GP gives each one a calibrated probability that a downstream tracker or alert system can act on.

To validate that GP probabilities are physically sensible on SOFT windows, Table 3 shows mean detection probability stratified by z-score sub-bin. The GP assigns low probability to near-threshold-but-not-quite z-scores and near-certainty to windows just below the threshold, despite never seeing SOFT windows during training. This monotonic relationship is exactly what a calibrated detector should exhibit.

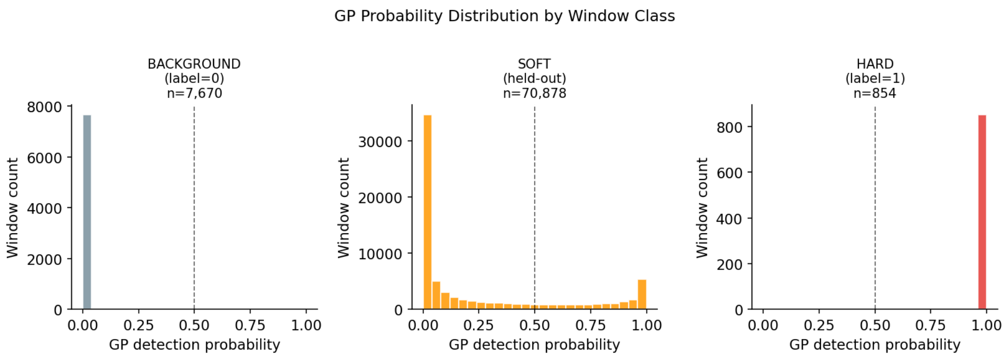


Figure 6: GP detection probability by window class. SOFT windows fill the uncertain middle range, showing calibrated uncertainty rather than overconfident predictions.

Table 3: SOFT-window GP detection probability stratified by z-score sub-bin. The model never sees these windows during training, yet probabilities scale monotonically with z.

z-score bin	Count	Mean GP prob	Frac. above 0.5
$z \in [3.0, 3.5)$	54,740	0.077	0.2%
$z \in [3.5, 4.0)$	16,138	0.836	99.8%

4.5 Pass-Level Aggregation

Threshold's F1=1.0 on pass-level detection is tautological, since positive passes are defined by having at least one window with $z \geq 4.0$, and the threshold fires on exactly those windows. The meaningful question is whether GP aggregation supports principled downstream tuning. Because the GP produces calibrated per-window probabilities, downstream systems can re-threshold the pass-level median without touching the underlying detector, trading precision and recall to match the cost structure of the application.

Each satellite overpass produces a sequence of 5–30 consecutive windows (∼2.9 s spacing). Aggregating window-level GP probabilities to a single pass-level decision via the median yields Table 4.

Table 4: Pass-level detection on the test split (5,526 passes, 341 positive).

Method	Precision	Recall	F1	False Positives
Threshold (any window ≥ 4.0)	1.000	1.000	1.000	0
GP median > 0.5	1.000	0.909	**0.952**	0
GP max > 0.5	1.000	1.000	1.000	0

Both aggregation methods achieve zero false positives. GP median is more conservative and misses passes where only 1–2 windows exceed 0.5, suggesting genuine borderline cases. The key advantage of GP aggregation over the threshold is that it works on calibrated per-window probabilities, so downstream systems can tune the aggregation threshold to trade precision and recall in a principled way without touching the underlying detector.

5 Conclusion

This project replaced the classical z-score threshold from the ECE 257B Starlink detection pipeline with a physics-informed Sparse Variational GP. The key contributions are:

1. **Calibration.** ECE $\sim 700\times$ lower than the threshold detector, showing the GP's probability outputs are trustworthy.
2. **Additive kernel design.** The ablation confirms this is the critical structural choice; a single RBF is $5\times$ worse in ECE despite identical ranking performance.

3. **Physics-informed features.** Adding angular velocity $\dot{\theta}$ activates the geometric kernel block. ARD lengthscales confirm the model learned to use this feature, with $\ell_{\dot{\theta}} = 2.19$ vs. $\ell_{\text{elev}} = 2.26$.
4. **Soft window coverage.** 70,878 previously discarded ambiguous windows now receive calibrated probabilities, enabling track-before-detect applications.

Limitations. Several caveats apply to the results above:

- No ground truth for SOFT windows; the z-bin stratification provides indirect evidence but not validation against an independent oracle. Future work could cross-validate with a second SDR at a different location.
- Dataset coverage is limited to high-elevation passes (49–90 degrees) because the pipeline stores only the highest-overhead satellite at each timestamp. A future campaign should record all visible satellites and include low-elevation geometry to fully stress the geometric kernel.
- The kernel structure is hand-designed based on physical priors. Deep kernel learning, which learns the feature representation jointly, could potentially improve ECE further.

Future work. Beyond these limitations, the calibrated GP probabilities enable track-before-detect applications over multi-pass time windows, which is the natural next step for operational deployment.

Code and GitHub

All code, notebooks, and results are available at:

`https://github.com/akshatmadrock/starlink-gp-detector`

Notebook 01 (EDA) generates exploratory figures and feature correlation analysis. Notebook 02 (Baselines) trains threshold, logistic regression, and MLP, and produces Table 2. Notebook 03 (SVGP) trains the GP, plots the reliability diagram, and generates the ECE comparison. Notebook 04 (Ablations) loads `results/ablation.json` and produces Figure 4. The enhancements pipeline (`scripts/enhancements.py`) generates Figures 5–6 and the lengthscale table. Raw SDR data (approximately 500 GB of IQ recordings) is not included; all experiments run on the processed features in the SQLite database `detections.db`, which is also excluded from the repo due to size. Intermediate results (metrics, predictions, figures) are committed to `results/`.

Team Member Contribution

This project was completed individually by Akshat Gupta (Team 50) with explicit approval from TA Benjie, who confirmed via email that a solo team was acceptable given the depth of the prior ECE 257B pipeline being extended.

Akshat Gupta: all contributions, including data collection and labeling (ECE 257B pipeline), feature engineering, SVGP model development, hyperparameter sweep, ablation study, angular velocity and ARD extensions, notebook authoring, and report writing.

References

[1] A. Gupta, "Passive Starlink Satellite Detection via Ku-Band SDR," ECE 257B Course Project Report, UC San Diego, June 2025. Available: `https://github.com/ece257b/starlink-passive-detection`

[2] J. Hensman, A. Matthews, and Z. Ghahramani, "Scalable variational Gaussian process classification," in *Proc. AISTATS*, 2015.

[3] C. E. Rasmussen and C. K. I. Williams, *Gaussian Processes for Machine Learning*. MIT Press, 2006.

[4] M. Titsias, "Variational learning of inducing variables in sparse Gaussian processes," in *Proc. AISTATS*, 2009.

[5] D. Milios, R. Camoriano, P. Michiardi, L. Rosasco, and M. Filippone, "Dirichlet-based Gaussian processes for large-scale calibrated classification," in *Proc. NeurIPS*, 2018.

[6] C. Williams and D. Barber, "Bayesian classification with Gaussian processes," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 12, 1998.

[7] M. Bhatt *et al.*, "Passive RF monitoring for LEO satellite constellations," *IEEE Access*, 2023.

[8] M. Neinavaie and Z. Kassas, "Acquisition, Doppler tracking, and positioning with Starlink LEO signals," *IEEE Trans. Aerosp. Electron. Syst.*, vol. 58, no. 4, 2022.

[9] D. Duvenaud *et al.*, "Structure discovery in nonparametric regression through compositional kernel search," in *Proc. ICML*, 2013.

[10] A. Solin and S. Särkkä, "Hilbert space methods for reduced-rank Gaussian process regression," *Statistics and Computing*, vol. 30, 2020.

[11] M. P. Naeini, G. Cooper, and M. Hauskrecht, "Obtaining well calibrated probabilities using Bayesian binning," in *Proc. AAAI*, 2015.

[12] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. ICML*, 2017.

Appendix: Teaching Evaluation Screenshot

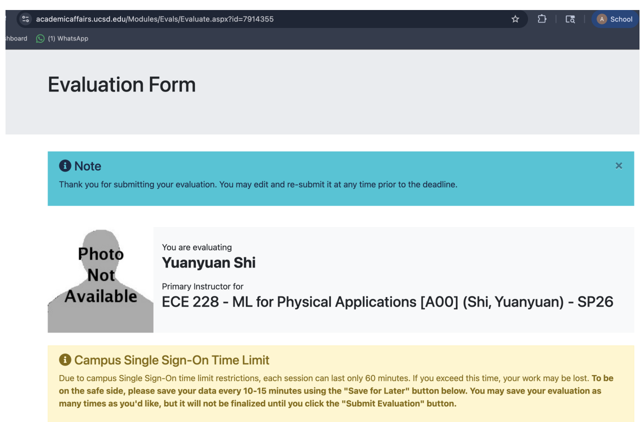


Figure 7: Teaching evaluation submission confirmation (5-point bonus).